Techfest 2026-27

इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY
सत्यमेव जयते

Security of Drones

Overview:

PUSHPAK Grand Challenge 2026: Towards Drone Excellence

A National Innovation Challenge under the PUSHPAK: National Mission on Drone Technology.

Funded by the **Ministry of Electronics and Information Technology (MeitY), Government of India.**

Hosted by **IIT Bombay.**

In collaboration with **VJTI Mumbai – Veermata Jijabai Technological Institute, Maharashtra (VJTI)**.

India is entering a new era of autonomous aerial systems, intelligent mobility, and drone-enabled digital infrastructure. The PUSHPAK Grand Challenge 2026 is a national platform created to identify and accelerate high-impact engineering solutions that can address real-world problems through drones, AI, robotics, sensing, communication systems, and autonomous technologies.

This is not a conventional hackathon or idea competition. The Grand Challenge is designed to take promising technical concepts through evaluation, mentoring, prototyping, testing, and deployment-oriented development. Selected teams will work with researchers, domain experts, and industry mentors to transform innovative ideas into functional, field-ready prototypes.

Grand Challenge 3, presented by VJTI at IIT Bombay Techfest 2026-27, is titled "**Security of Drones**."

Guiding Professors / Mentors

Grand Challenge 3 (Security of Drones) will be guided by the following faculty mentors from the collaborating institutes:



Prof. Arnab Maity

IIT Bombay – Aerospace Engineering

Prof. Arnab Maity is an accomplished aerospace researcher and faculty member at IIT Bombay with strong expertise in dynamics, guidance, navigation and control of aerospace and autonomous systems. His key strength lies in combining rigorous theoretical foundations in control, estimation, optimization and system dynamics with their application to complex real-world aerospace problems. His research encompasses autonomous flight systems, UAV and drone technologies, guidance and navigation algorithms, adaptive and optimal control, and aerospace vehicle and propulsion-system control. With a strong analytical and systems-oriented approach, Prof. Maity contributes to the development of robust, intelligent and reliable autonomous platforms, particularly for applications where safety, precision and real-time decision-making are critical. His expertise makes him well positioned to contribute to advanced drone technology initiatives involving autonomous navigation, flight control, collision avoidance, multi-agent systems and next-generation UAV applications, while bridging fundamental research with technology development and mission-oriented outcomes.



Dr. Faruk Kazi

Professor, VJTI Mumbai; Director – Research and Development Cell, University of Mumbai

Dr Faruk Kazi received his Ph.D. degree from IIT Bombay in 2009. He is currently working as Professor at VJTI, Mumbai. He is also Director – Research and Development Cell, University of Mumbai. He is a senior member of IEEE and is a recipient of AICTE-VISVESVARAYA Best Teacher National Award-2020. He has worked as visiting research scholar at Swiss Federal Institute of Technology, EPFL, Switzerland, MIT USA, and Tufts University, USA. He has extensively worked in the areas of digital architecture and cyber security of critical energy infrastructure. He has contributed in developing resilience framework for smart grid and other smart city verticals. ICS-CERT & OEMs have released around 80 CVEs and worldwide advisories based on vulnerabilities identified by his research team. He is listed on "Schneider Electric Wall of Thanks-2021" for helping in securing Schneider Electric products. He has published more than 100 research papers in IEEE Journals & Conferences in the domain of smart grid, cyber-physical systems and critical infrastructure and embedded security. His contribution to Critical Infrastructure Cyber Security spans the complete research-to-deployment and policy-to-capacity-building spectrum—from identifying vulnerabilities in real-world ICS/OT products and developing attack-analysis methodologies, to strengthening cyber resilience of power transmission systems through Digital Twins, addressing GPS/NavIC security in smart grids, contributing to national critical-information-infrastructure frameworks, and building cybersecurity capabilities.

What Happens After the Grand Finale?

The journey does not end with the finale.

Prototype Development Support (Build & Test Phase)

Selected winning teams (upto 5 teams) will be invited to participate in a dedicated Prototype Development Support Programme, beginning December 2026 onwards. This post-challenge phase is specifically

designed to bridge the gap between academic/start-ups innovation and real-world deployment.

Build the Future of India's Drone Ecosystem

If you have an idea that can redefine autonomous flight, intelligent sensing, aerial mobility, infrastructure automation, and the cybersecurity of drones, this is your opportunity.

Design. Build. Test. Demonstrate.

Join the **PUSHPAK Grand Challenge 2026** and help create the next generation of indigenous drone technologies for India and the world.

Challenge Questions

Objective:

There are two Objectives under this Grand Challenge "Security of Drones"

Objective 1 - Drone Forensics Toolkit Development

Objective 2 - Drone Intrusion Detection System (Drone IDS)

Note: Teams can participate in either of the Objectives which will be evaluated as per the details provided. At the end of the challenge, up to 5 winning teams will be selected for Grand challenge 3. These teams will be considered for further support, funding, mentoring, and access to resources for building and testing their designs in the next phase.

Objective 1: Drone Forensics Toolkit Development

1.1 Objective

The objective of this Grand Challenge is to design, develop, and validate an indigenous Drone Forensic Toolkit capable of acquiring, preserving, analyzing, and reporting digital evidence from Unmanned Aerial Vehicles (UAVs) and their associated components in a forensically sound manner. The toolkit should support evidence acquisition from onboard storage, flight controllers, remote

controllers, removable media, companion mobile applications, and other relevant data sources while ensuring integrity through cryptographic hashing. The solution should enable automated data extraction, timeline reconstruction, flight path analysis, metadata extraction, and comprehensive forensic reporting, and should be scalable, extensible, and suitable for deployment by law enforcement agencies, forensic laboratories, regulatory authorities, and defence organizations for investigation of drone-related incidents.

1.2 Problem Statement

Despite the growing importance of drone forensics, the investigation process remains challenging due to diverse hardware architectures, proprietary data formats, encrypted storage, vendor-specific communication protocols, volatile memory, and the absence of standardized forensic acquisition and analysis methodologies. These challenges make it difficult to preserve evidence integrity and perform comprehensive forensic investigations across different UAV platforms.

The toolkit should support the acquisition of digital evidence from onboard storage, removable storage media, flight controllers, remote controllers. The proposed solution should incorporate secure evidence preservation mechanisms, cryptographic hashing for integrity verification, chain-of-custody management, automated evidence parsing, timeline reconstruction, and standardized forensic reporting suitable for investigative and legal proceedings. The toolkit should also demonstrate repeatable and verifiable forensic acquisition processes under controlled test conditions.

Drone Forensic Toolkit system should be modular, scalable, vendor-independent, and easily extensible to support emerging UAV technologies, new drone platforms, and evolving forensic requirements with minimal modifications. The ultimate goal is to establish an indigenous, reliable, and standardized drone forensic capability that can support law enforcement agencies, regulatory bodies, forensic laboratories, defense organizations, and cybersecurity professionals in conducting efficient and legally defensible drone investigations.

1.3 Evaluation Criteria

Criterion	Weight
Successful forensic acquisition from multiple drone platforms	20%

Evidence integrity using hashing and chain-of-custody	20%
Ability to recover logs, telemetry, GPS, media, and configuration files	20%
Completeness of forensic reporting	15%
Automation and usability of forensic toolkit	10%
Extensibility to new drone platforms	5%
Documentation and reproducibility	5%
Future integration potential	5%

Total: 100%

Final evaluation will be based on forensic acquisition, evidence preservation, repeatability, reporting quality, end-to-end investigation capability and viva during the Techfest evaluation.

1.4. Structure

The Grand Challenge will be conducted over approximately 4 months and will culminate in final presentations at IIT Bombay Techfest in December 2026.

Note: Teams can participate in either of the Objectives which will be evaluated as per the details provided. At the end of the challenge, up to 5 winning teams will be selected for Grand challenge 3. These teams will be considered for further support, funding, mentoring, and access to resources for building and testing their designs in the next phase.

Stage 1: Problem Statement Announcement and Preliminary Design

The Grand Challenge will begin with the announcement of the problem statement. Teams will be given one month to prepare and submit a preliminary design for Drone forensic toolkit.

The preliminary submission should include: Drone Forensic Toolkit architecture, forensic methodology, supported UAV platforms, digital evidence acquisition workflow, hardware acquisition interfaces, evidence preservation mechanisms, chain-of-custody methodology, cryptographic hashing approach, metadata

extraction strategy, reporting framework, software architecture, risk assessment, and validation methodology.

No financial support will be provided at this stage. The objective of this stage is to identify technically promising teams with strong and feasible solutions.

Stage 2: Drone Forensic Toolkit Detailed Design and Prototype Development

The organizing team will take approximately two weeks to evaluate preliminary designs and shortlist up to 15 teams. These teams will receive Rs. 1 lakh each to develop a complete, detailed design and prototype.

This phase will last approximately two months. Teams shall prepare a Drone Forensic Toolkit capable of performing physical and logical acquisition of digital evidence from supported UAV platforms. The toolkit shall include evidence acquisition modules, forensic parsers, evidence preservation mechanisms, cryptographic hashing, chain-of-custody management, evidence repository architecture, timeline reconstruction, and automated forensic reporting.

Stage 3: Drone Forensic Toolkit Detailed Design and Prototype Evaluation

After submission of Drone Forensic Toolkit detailed design and prototype reports, the organizing team will take approximately two weeks to evaluate the submissions. Up to 10 teams will be shortlisted for final presentation during IIT Bombay Techfest in December 2026.

During this stage, the evaluation committee may provide brief feedback to the shortlisted teams. Teams will be encouraged to improve their design, prototype, documentation, and presentation before the final round. Travel or participation support may be provided to the shortlisted finalist teams, subject to the approved budget norms.

Final Presentation: Techfest Evaluation

The final round will be conducted during IIT Bombay Techfest in December 2026. Finalist teams will present their detailed design and prototype before the evaluation committee and respond to technical questions.

The final evaluation will be based on:

- Detailed design, prototype and demonstration.

- Supported technical evidence.
- Final presentation.
- Viva and technical defence.
- Feasibility of implementation in the next phase.

Up to 5 winning teams will be selected for Grand challenge 3. These winners will receive awards or recognition grants and will be considered for further financial support, technical mentoring, and access to resources for prototype development and testing with IIT Bombay support.

Note: Teams can participate in either of the Objectives which will be evaluated as per the details provided. At the end of the challenge, up to 5 winning teams will be selected for Grand challenge 3. These teams will be considered for further support, funding, mentoring, and access to resources for building and testing their designs in the next phase.

1.6 Expected Target Audience

The Grand Challenge is intended for participants with expertise in digital forensics, cybersecurity, embedded systems, and UAV technologies that are capable of developing innovative forensic solutions for drone investigations. The Grand Challenge is open to academic institutions, research laboratories, startups, MSMEs, defence organizations, and industry consortia possessing expertise in digital forensics, cybersecurity, embedded systems, and UAV technologies.

The challenge aims to foster indigenous development of advanced drone forensic capabilities suitable for law enforcement agencies, forensic laboratories, regulatory bodies, and defence organizations.

Project staff, research staff, consultants, interns, and other personnel directly engaged with the PUSHPAK Project, and Drone Centre, or the organizing/host institutions in connection with the Grand Challenge are NOT eligible to participate in the Grand Challenge.

This restriction applies whether such personnel participate individually or as part of a team.Any team found to include an ineligible participant may be disqualified at any stage of the Grand Challenge, including after selection or announcement of results. The organizers reserve the right to verify the eligibility of participants at any stage and their decision in this regard shall be final.

Objective 2 Drone Intrusion Detection System (Drone IDS)

2.1 Objective

The objective of this Grand Challenge is to encourage the development of innovative, indigenous, and deployable Drone Intrusion Detection Systems (Drone IDS) that provide real-time cyber threat detection and monitoring for UAV platforms. The challenge aims to foster solutions capable of detecting communication, navigation, firmware, and control system attacks with high accuracy, low latency, and minimal resource overhead, while enabling seamless integration into existing and future UAV platforms for civilian, industrial, and defence applications.

2.2 Problem Statement

The objective of this challenge is to design, develop, and validate an indigenous Drone Intrusion Detection System (Drone IDS) capable of continuously monitoring UAV systems and detecting cyber threats in real time. The Drone IDS should operate autonomously with minimal computational and power overhead, making it suitable for deployment on resource-constrained UAV platforms. It should provide real-time threat detection, event logging, attack classification, secure alert generation, and comprehensive reporting to support timely response and post-incident analysis. The solution should be modular, scalable, and interoperable with a wide range of UAV platforms and communication protocols, enabling seamless integration into existing and future indigenous drone ecosystems.

The Drone IDS system should demonstrate reliable detection of representative cyber threats under controlled test conditions and be suitable for future deployment in civilian, industrial, critical infrastructure, public safety, and defence applications. The challenge focuses on developing an onboard cyber defence capability that enhances the resilience, safety, security, and trustworthiness of next-generation UAV platforms.

2.3 Evaluation Criteria

Criterion	**Weight**
Detection accuracy across attack scenarios	20%
False Positive Rate (FPR)	20%
Distance Covered	10%

Detection latency	10%
Coverage of multiple attack vectors	15%
Computational efficiency	10%
Ease of integration	5%
Documentation and validation	5%
Future deployment potential	5%

Total: 100%

Final evaluation will be based on live demonstrations, and viva during the Techfest evaluation.

2.4 Structure

The Grand Challenge will be conducted over approximately 4 months and will culminate in final presentations at IIT Bombay Techfest in December 2026.

Note: Team can participate in either or both Objectives. At the end of the challenge, up to 5 winning teams will be selected. These teams will be considered for further support, funding, mentoring, and access to resources for building and testing their designs in the next phase.

Stage 1: Problem Statement Announcement and Preliminary Design

The Grand Challenge will begin with the announcement of the problem statement. Teams will be given one month to prepare and submit a preliminary design.

Teams will submit their proposed Drone Intrusion Detection System (Drone IDS) architecture, threat catalogue, attack models, detection methodology, dataset strategy, feature extraction approach, logging framework, sensor selection, compute platform selection, system integration methodology, benchmarking strategy, and validation approach.

No financial support will be provided at this stage. The objective of this stage is to identify technically promising teams for detailed design support.

Stage 2: Drone IDS System detail design and prototype Development

The organizing team will take approximately two weeks to evaluate preliminary designs and shortlist up to 15 teams. These teams will receive Rs. 1 lakh each to develop a complete detailed design and a functional Drone IDS prototype.

During this stage the teams will develop a working Drone Intrusion Detection System capable of monitoring UAV communication channels, onboard sensors, navigation data, telemetry, flight control commands, and firmware integrity for real-time cyberattack detection, and other security-relevant system parameters. The prototype shall be validated against representative attack scenarios (including GPS spoofing, MAVLink communication anomalies, command injection, telemetry manipulation, denial-of-service attacks, and firmware integrity violations).

Stage 3: Drone IDS System detail design and prototype Evaluation

After submission of the detailed design and Drone IDS prototype, the organizing team will take approximately two weeks to evaluate the submissions. Up to 10 teams will be shortlisted for final presentation during IIT Bombay Techfest in December 2026.

During this stage, the evaluation committee may provide brief feedback to the shortlisted teams. Teams will be encouraged to improve their design, prototype, documentation, and presentation before the final round. Travel or participation support may be provided to the shortlisted finalist teams, subject to the approved budget norms.

Final Presentation: Techfest Evaluation

The final round will be conducted during IIT Bombay Techfest in December 2026. Finalist teams will present their detailed design, Drone IDS prototype, and demonstration before the evaluation committee and respond to technical questions.

The final evaluation will be based on:

- Detailed design, functional prototype, and demonstration.
- Supported technical evidence.
- Final presentation.
- Viva and technical defence.
- Feasibility of implementation in the next phase.

Up to 5 winning teams will be selected for Grand challenge 3. These winners will receive awards or recognition grants and will be considered for further financial

support, technical mentoring, and access to resources for prototype development and testing with IIT Bombay support.

2.6 Expected Target Audience

The Grand Challenge is intended for participants with expertise in cybersecurity, UAV technologies, embedded systems, artificial intelligence, and networking that are capable of developing advanced cyber-defense solutions for drone platforms. The challenge encourages participation from academic institutions, faculty-led research groups, drone and cybersecurity startups, MSMEs, defence organizations, R&D laboratories, and academia–industry consortia. Preference will be given to teams with demonstrated expertise in UAV security (Intrusion Detection Systems).

Project staff, research staff, consultants, interns, and other personnel directly engaged with the PUSHPAK Project, and Drone Centre, or the organizing/host institutions in connection with the Grand Challenge are NOT eligible to participate in the Grand Challenge.

This restriction applies whether such personnel participate individually or as part of a team. Any team found to include an ineligible participant may be disqualified at any stage of the Grand Challenge, including after selection or announcement of results. The organizers reserve the right to verify the eligibility of participants at any stage and their decision in this regard shall be final.

Timeline and Key Dates:

Stage	Timeline(s)	Participation Brief	Prize / Reward / Support
Grand Challenge Announcement	22 August 2026	PUSHPAK Grand Challenge 2026 opens for registrations across India. Teams of up to 5 members may register and participate.	None

Stage 1: Preliminary Design Verification	Work Time: 22 August – 27 September 2026 Submission Deadline: On or Before 27 September 2026 Results Announcement: 02 October 2026	All registered teams are eligible. Submit a 6–8 page technical proposal, software architecture, working proof-of-concept simulation, source code, installation instructions and demonstration video. (Upto 15 teams shall qualify for next stage)	INR 1 Lakh per qualifying team after the announcement of results of Stage 1
Stage 2: Technical Challenge	Work Time: 03 October – 02 December 2026 Submission Deadline: On or Before 02 December 2026 Results Announcement: 06 December 2026	Shortlisted teams develop a detailed engineering design, simulation results, subsystem architecture, implementation methodology, technology stack, risk assessment, and prototype development roadmap. (Upto 10 Teams shall qualify for next stage)	Domestic Travel & Accommodation support as per guidelines of IIT Bombay to be shared at a later stage for qualifying team
Stage 3: Grand Challenge Finale	16 – 18 December 2026	Finalist teams present and demonstrate their solution at IIT Bombay Techfest. (Upto 5 winning teams get a chance to enter the Post-Grand Challenge Stage)	Zenith – INR 3 Lakh, Horizon – INR 2.5 Lakhs, Vector – INR 2 Lakhs, Ascent (x2) – INR 1.5 Lakh, for the winning team after the announcement of results at the TechFest

Registration & Submission

- Participants must register for the PUSHPAK Grand Challenge on the official Techfest website: techfest.org > Competitions > PUSHPAK Grand Challenge > Register at www.techfest.org/competitions

- All submissions (Stage 1 and Stage 2) are to be made via email to [pushpak_gc2026@aero.iitb.ac.in] — no submission links are used; every stage is handled entirely through email correspondence.

Prize Money

Stage 1 (Preliminary Design Verification): INR 1 Lakh per qualifying team after the announcement of results of Stage 1.

Stage 2 (Technical Challenge): Domestic Travel & Accommodation support as per guidelines of IIT Bombay to be shared at a later stage for qualifying teams.

Stage 3 (Grand Challenge Finale):

Zenith: INR 3 Lakh

Horizon: INR 2.5 Lakhs

Vector: INR 2 Lakhs

Ascent (x2): INR 1.5 Lakh

for winning teams after the announcement of results at the Techfest.

Note: At the end of the challenge, up to 5 winning teams will be selected. These teams will be considered for further support, funding, mentoring, and access to resources for building and testing their designs in the next phase.

The process regarding the disbursement of prize money will be communicated at a later stage. Participating teams are requested to regularly check the official website and their registered email accounts for updates.

Disclaimer

Organizer's Rights: The organizers reserve the right to modify, extend, postpone, suspend, or cancel the Grand Challenge, or any stage thereof, at their

discretion. Any such changes will be communicated through the official communication channels of the Grand Challenge.

Eligibility & Participation: Participation in the Grand Challenge is subject to the eligibility criteria, rules, guidelines, and submission requirements specified by the organizers. Submission of an entry does not guarantee selection for any subsequent stage or award.

Evaluation: All submissions will be evaluated by the designated evaluation committee based on the criteria specified for the respective stage. The decision of the evaluation committee shall be final and binding, and no correspondence regarding the evaluation or ranking will be entertained.

Technical Responsibility: Participants are solely responsible for the accuracy, completeness, originality, technical feasibility, safety, and performance of their submissions. Selection or recognition of a submission by the organizers does not constitute certification, approval, or endorsement of the proposed technology for operational deployment.

Safety & Compliance: Participants shall ensure that their proposed solutions and demonstrations comply with all applicable laws, regulations, standards, aviation requirements, safety protocols, and permissions applicable in India. Participants shall be responsible for obtaining any permissions, approvals, or clearances required for testing or demonstration.

Intellectual Property: Participants shall ensure that their submissions do not infringe upon any third-party intellectual property rights. Participants retain ownership of their intellectual property unless otherwise specifically agreed under the terms of the Grand Challenge. Any use, disclosure, licensing, or transfer of intellectual property arising from the Grand Challenge shall be governed by the applicable terms and agreements communicated by the organizers.

Confidentiality: Participants should clearly identify any information submitted that they consider confidential or proprietary. Participants are responsible for ensuring that they do not disclose information belonging to third parties without appropriate authorization.

Prizes & Recognition: Prizes, awards, certificates, or other forms of recognition are subject to fulfillment of the applicable eligibility and evaluation requirements. The organizers reserve the right to withhold or modify an award where no submission meets the required standards.

Expenses: Unless expressly stated otherwise, participation in the Grand Challenge, including development, testing, travel, accommodation, equipment, and demonstration-related expenses, shall be borne by the participating teams.

Liability: The organizers, host institutions, funding agencies, partners, mentors, evaluators, and their respective personnel shall not be liable for any loss, damage, injury, delay, technical failure, or other consequence arising from participation in the Grand Challenge, except to the extent required by applicable law.

Force Majeure: The organizers shall not be responsible for any delay, modification, suspension, or cancellation resulting from circumstances beyond their reasonable control, including natural disasters, technical failures, regulatory changes, government directives, or other unforeseen events.

Acceptance of Terms: By registering for or participating in the Grand Challenge, participants acknowledge that they have read, understood, and agreed to abide by the applicable rules, guidelines, terms, and conditions of the Grand Challenge.